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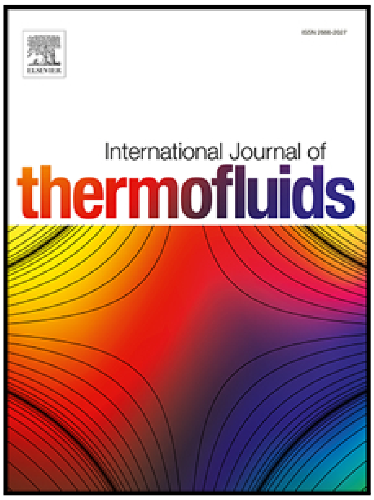

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Computational simulation of Casson hybrid nanofluid flow with Rosseland approximation and uneven heat source/sink

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Abstract

This article candidly presents the magnetohydrodynamics Casson hybrid nanofluid flow over a stretching surface. In the present study, we added a nonuniform heat source or sink and non-linear thermal radiation. We considered Al_2O_3 and copper nanoparticles to have antibacterial and antiviral properties without any harmful impacts and used water as the host fluid. We simplified the governing flow equations by using suitable self-similarity variables, which are used to convert PDEs to ODEs. The mathematical equations are numerically solved by using the bvp5c technique in the MATLAB software. Additionally, with higher values of the magnetic field and Casson fluid parameters the velocity profile decreased. The temperature profile is enhanced by increasing the magnetic field and thermal radiation parameters. Increasing the Casson fluid and radiation parameters enhances the skin friction and Nusselt number profiles. Alumina nanoparticles find applications in cosmetic fillers, polishing materials, catalyst carriers, analytical reagents. Copper nanoparticles have high electrical conductivity, which has many uses in electrical circuits and biosensors.

Keywords: Casson fluid, heat source or sink, hybrid nanofluid and Roseland thermal radiation approximation.

Nomenclature

u, v	Velocity components in x and y directions (ms^{-1})	Pr	Prandtl number
T	Fluid temperature (K)	Ec	Eckert number
u_w	Velocity	Re_x	Local Reynolds number
V	Fluid Velocity	Nu_x	Nusselt number
u_∞	Free stream velocity	Br	Brinkmann Number
T_w	Fluid temperature	*Greek symbols*	
T_∞	Temperature in free stream	μ_B	Plastic dynamic viscosity of the non-Newtonian fluid
K	Thermal velocity slip factor	ρ_{hnf}	Density of hybrid nanofluid ($kg\ m^{-3}$)
$(c_p)_{hnf}$	specific heat at a constant pressure of hybrid nanofluid ($Jkg^{-1}K^{-1}$)	ψ	Stream function
M	Magnetic parameter	σ^*	Stefan–Boltzmann constant
R	Radiation parameter	υ_{hnf}	Kinematic viscosity of hybrid nanofluid
q_r	Radiative heat flux	υ_f	Kinematic viscosity of fluid
B_0	Uniform Magnetic field	μ_{hnf}	Dynamic viscosity of hybrid nanofluid
k_{hnf}	Thermal conductivity of hybrid nanofluid ($w\ m^{-1}k^{-1}$)	θ	Dimensionless temperature
V_w	Wall mass transfer	θ_w	Temperature ratio parameter
q'''	Non-uniform heat source/sink	γ	Casson Parameter
A^*	Coefficient of space dependent heat source/sink	η	Similarity variable
B^*	Coefficient of temperature dependent heat source/sink	ρ	Density of the fluid
k^*	Rosseland mean absorption coefficient	*Subscripts*	
f'	Dimensionless velocity	w	Conditions at the wall
C_f	Skin friction coefficient	∞	Ambient condition
		Superscript	
		$'$	Differentiation with respect to η

1. Introduction

Nanotechnology expanding hastily its wings multifold in is efforts in science and technology, by which new materials on the nanoscale level can be manufactured. It refers to the development or application of particles with dimension that fall into the nanometer range ie., one billionth of a meter (1 nm = 10^{-9}m). The term 'nano' comes from the Greek word, means dwarf. "Top down" and "Bottom up" are two potential approaches of nanotechnology. The "top down" endeavor involves conversion of big size materials into nanometer size by making use of different mechanical techniques such as grinding, milling, so forth. On the other hand, "bottom-up" methodology was follow-on from the basic theory of biology like self-gathering and self-association. By slow collection of atom by atom or molecules by molecules show the way to engineering of large. It comprises chemical synthesis, self- gathering and positional assembly. A nanofluid is a fluid containing nanoparticles (less than 100 nm-diameters) distributed in a conventional base liquid (H_2O, toluene, ethylene glycol etc.), which generates desired and elevated thermal conductivity and used in many thermal systems such as biomedical devices, reduction of heat in electronic components, nuclear reactors and so forth. In fact, the word nanofluid was coined by Choi [1] in 1995. The surface area of the nanoparticles is high than volume ratio and has exclusively physical and chemical properties which give rise them as antimicrobial agents [2, 3]. Some relevant discussions on nanoliquids can be pinpointed in Suneetha et al. [4], Ramesh et al. [5], and Mohanty et al. [6-9], Dharmaiah et al. [10-11], Ali et al. [12].

Concurrent with the use of these fluids, researchers also made use of a new sort of nanofluid known as a "hybrid" nanofluid, which is made by suspending nanoparticles of more than one type in a base fluid. These hybrid nanofluids are more stable and suitable than the mono nanofluids. Hybrid nanofluid is considered as the heat transfer fluid which has tremendous applications in areas like: naval structure, solar heating, microelectronics, medical, micro fluidics, transportation, heat exchanger, car radiators, nuclear plant etc. Many researchers [13-20] carried hybrid nano fluids on their work in different geometries.

Here, we study the effect of commercially obtainable silver (Ag) and copper (Cu) nanoparticles. Silver nanoparticles are helpful as an antiviral agents, photo and radio sensitizers, and therapeutic agent in anticancer treatment etc. for treating leukemia, lung carcinoma, and breast cancer so forth [21]. Also, Ag NPs use for antimicrobial coatings, key boards, wound dressings, and biomedical devices where silver ions are released continuously at a small amount which give protection from bacteria. Cu NPs behave as antibacterial and antifungal agents [22]. In addition, they are used as anti-inflammatory, anti-oxidative and anti-ulcer agents [23-24]. Cu NPs have high electrical conductivity which has many uses in electrical circuits, biosensors and sintering additives. Also used in medicine to prevent skin photosensitivity are used in cream and skin care spray and lotion.

Newtonian and non-Newtonian fluids are the two types of fluids that can be separated from one another in nature according to their viscosity. In Newtonian fluids, the shear stress acts in a manner that is linearly proportional to the shear rate, provided that there is no yield stress.
H_2O, motor oil with low concentration, alcohol, and air are examples of Newtonian fluids that are commonly encountered. It is difficult to predict the viscosity of most naturally occurring fluids since they are not Newtonian. This is because their viscosity changes as the strain rate does. The non-linear rheological properties of these liquids play a significant role in different industries such as milk production, pharmaceuticals, fiber technology, and energy storage. Extremely non-linear components are introduced into the governing differential equations when a non-Newtonian fluid model is being modelled.
There are numerous fluid models that have been presented as a result, including Casson, Maxwell, Jeffrey, Oldroyd-A, Eyring-Powell, Oldroyd-B, Carreau and others. Although there are several models for suspensions in everyday life, the most important one is the plastic fluid Casson model [25-28]. This model has applications in many different areas, including processing of food, metallurgy, drilling activities, and bio-engineering. A shear thinning fluid, Casson fluid has an incalculable viscosity at zero shear rates. Imagine a solid that acts like elastic. As the shear stress increases, this model degenerates into a viscous fluid. It is important to note that this particular Casson model takes into account the rheological effects of a variety of other fluids as well, including syrups, cosmetics, physiological fluids, foams, and so on.

Heat transfer owed to radiation has applications in nuclear power plants, aircrafts, gas turbines, space vehicles and satellites. The linear radiation is not appropriate for systems with high temperature variations because in Rosseland approximation linearization the dimensionless parameter effective Prandtl number [29-32] is only used, while in the nonlinear approximation, the radiation, Prandtl number and the temperature ratio parameters are used in the governing problem. Pantokratoras [33] was the first person who explored the influence of linear, nonlinear radiation on a vertical plate with anew parameter called as film radiation parameter. The advanced science and technology are gratified to nonlinear radiation as such its impact and intensity help to design devices at nano level. The nonlinear radiative energy of a nano fluid is absorbed by the Nano sized particles which are in smaller size than de Broglie wave length. As a result, nanoparticles amplify the radiative features of fluids, which initiates in enhancing the effective absorption of solar collectors. The former person who inspected the nano particles absorbing radiation was Hunt [34]. Similar studies can be seen in the mentioned references [35-41]. Radiative features also assist for hydrothermal relations of nanofluid flow over different geometries [42, 43].

As per the author knowledge these kinds of model have not been examined before. Keeping these in mind, in the current work we employed heat transfer analysis on Casson hybrid nanofluid flow over a stretching surface with the presence of magnetohydrodynamics and uneven heat source or sink. The non-linear thermal radiation is preferred to see the temperature alterations in this model. The present mathematical model is applied to two different nano particles like Al_2O_3 and Cu along with Water as a base fluid. Solving the ordinary differential equations by using the bvp5c technique in MATLAB solver. Results and discussion section, graphs provided for various physical importance active parameters. As a consequence, researchers are certain believe the latest research is individual will have an important effect in the fields of both engineering and medical has the potential to inspire upcoming researchers.

2. Mathematical conceptualization

We deliberated the scenario on nanoliquid flow of two-dimensional incompressible viscous fluid with magnetohydrodynamics, thermal radiation and uneven source of heat or sink. Two different types of nanoparticles i.e., Al_2O_3 and copper (Cu) and water as a host fluid are pondered. The uneven source of heat or sink and radiation are highlighted specially.

The rheological expression for an incompressible flow via a Casson-type fluid follows [43-46]:

$$\tau_{ij}^{*}=\begin{cases}2\left(\mu_b^{*}+\dfrac{p_y^{*}}{\sqrt{2\pi^{*}}}\right)e_{ij}^{*}, & \pi^{*}>\pi_c^{*}\\ 2\left(2\mu_b^{*}+\dfrac{p_y^{*}}{\sqrt{2\pi_c^{*}}}\right)e_{ij}^{*}, & \pi^{*}\le\pi_c^{*}\end{cases}$$

Here $\pi^{*}=\left(e_{ij},\ e_{ij}\right)$ product of element of the deformation rate, p_y^{*} represents a yield stress, μ_b^{*} represents a plastic dynamic viscosity of the Casson fluid model, π_c^{*} represents a critical value of this item according to the Casson fluid model,.

The above equation takes the following form when $\pi^{*}\le\pi_c^{*}$: $\tau_{ij}^{*}=2\mu_b^{*}\left(1+\dfrac{1}{\gamma}\right)e_{ij}^{*}$, where the Casson fluid parameter is $\gamma = \frac{\mu_b^* \sqrt{2\pi_c^*}}{p_y^*}$.

All the physical description and active physical factors on the stated model and flow geometry of the hybrid nanofluid are established in Fig.1 and the flow expressions are as follows:

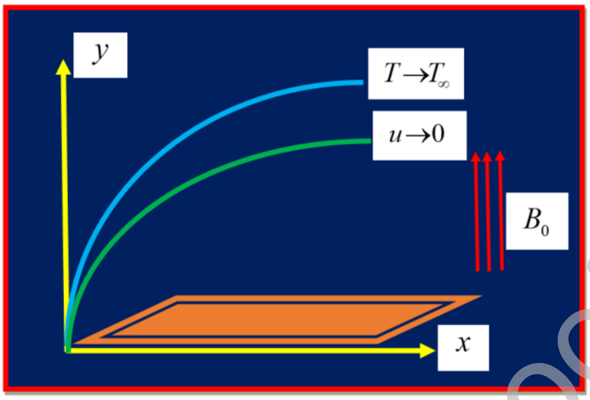

Fig. 1 Flow sketch.

$$\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} = 0, \tag{1}$$

$$u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y} = \upsilon_{hnf}\left(1 + \frac{1}{\gamma}\right)\frac{\partial^2 u}{\partial y^2} - \frac{1}{\rho_{hnf}}\sigma B_0^{\,2} u, \tag{2}$$

$$\left(u\frac{\partial T}{\partial x} + v\frac{\partial T}{\partial y}\right) = \frac{k_{hnf}}{\left(\rho c_p\right)_{hnf}}\frac{\partial^2 T}{\partial y^2} + \frac{\mu_{hnf}}{\left(\rho c_p\right)_{hnf}}\left(\frac{\partial u}{\partial y}\right)^2 + \frac{\sigma_{hnf} B_0^{\,2}}{\left(\rho c_p\right)_{hnf}} u^2 - \frac{1}{\left(\rho c_p\right)_{hnf}}\frac{\partial q_r}{\partial y} + \frac{1}{\left(\rho c_p\right)_{hnf}} q''', \tag{3}$$

The uneven heat source or sink, q''' is modeled as

$$q''' = \frac{k_{hnf} u_w}{x \upsilon_{hnf}}\left[f' A^* (T_w - T_\infty) + B^* (T - T_\infty)\right], \tag{4}$$

The space-dependent and heat source/sink properties are denoted by the letters A^* and B^*, respectively. The circumstances where the value of (A^*, B^*) is greater than zero correspond to the creation of internal heat, whereas the cases where (A^*, B^*) is less than zero relate to the reversal of internal heat.

The radiative heat flux q_r for optically thick boundary layer is

$$q_r = -\frac{4\sigma^*}{3k^*}\frac{\partial T^4}{\partial y}, \tag{5}$$

where σ^* represents the Stefan – Boltzmann constant, k^* denotes the Rosseland mean absorption coefficient. The temperature difference within the flow is assumed to be sufficiently small so that T^4 may be expressed as a linear function of T and T_∞ expanded by using Taylor's series:

$$T^4 = T_\infty^4 + 4T_\infty^3\left(T - T_\infty\right) + 6T_\infty^2\left(T - T_\infty\right)^2 + \dots$$

By neglecting the higher order terms beyond the first degree in $\left(T - T_\infty\right)$, we obtained:

$$T^4 = -3T_0^4 + 4T_0^3 T.$$

With Eq's (4) and (5) in Eq. (3), then it turns out to be as:

$$\left(u\frac{\partial T}{\partial x} + v\frac{\partial T}{\partial y}\right) = \frac{k_{hnf}}{\left(\rho c_p\right)_{hnf}}\frac{\partial^2 T}{\partial y^2} + \frac{\mu_{hnf}}{\left(\rho c_p\right)_{hnf}}\left(\frac{\partial u}{\partial y}\right)^2 + \frac{\sigma_{hnf} B_0^{\ 2}}{\left(\rho c_p\right)_{hnf}}u^2 - \frac{1}{\left(\rho c_p\right)_{hnf}}\frac{16\sigma^* T^3}{3k^*}\frac{\partial}{\partial y}\left(T^3\frac{\partial T}{\partial y}\right) + \frac{1}{\left(\rho c_p\right)_{hnf}}\frac{k_{hnf} u_w}{x\upsilon_{hnf}}\left[f' A^*(T_w - T_\infty) + B^*(T - T_\infty)\right], \quad (6)$$

The thermo physical values for the hybrid nanofluid are framed in (Table 1).

Table 1. Thermophysical properties of water $\left(H_2O\right)$, copper $\left(Cu\right)$, and Aluminium oxide $\left(Al_2O_3\right)$ Ramasekhar and Reddy [45]

Parameters	***ρ (kg / m³)***	***C_p (J / kgK)***	***k (W / mK)***	***σ (Ωm)⁻¹***	***Pr***
Water $\left(H_2O\right)$	997.1	4180	0.613	0.05	6.20
Copper $\left(Cu\right)$	8933	385	401	59.6×10^6	
Aluminium oxide $\left(Al_2O_3\right)$	3970	765	40	35×10^6	

The associated conditions at boundary are:

$$u = u_w,\ v = V_w,\ T = T_w \text{ at } y = 0,$$

$$u \to 0,\ T \to T_\infty \quad \text{as a } y \to \infty, \quad (7)$$

Where $V_w = \sqrt{\upsilon_f u_w / x}\ f(0)$, (8)

Eq.(8) be the species transfer at the surface of the wall with a velocity V_w, where $V_w > 0$ and $V_w < 0$ be the injection and suction.

Imparting the correspondence variables:

$$\eta = y\left(\frac{u_w}{v_f x}\right)^{\frac{1}{2}}, \varphi = \sqrt{v_f x u_w}\, f(\eta), T = \left[\theta_w\theta(\eta) - \theta(\eta) + 1\right]T_\infty,\ \theta = \frac{T - T_w}{T_w - T_\infty} \tag{9}$$

here η - similarity variable, φ - stream function and $\theta_w = \frac{T_w}{T_\infty}$.

$u = \partial\varphi/\partial y, v = -\partial\varphi/\partial x$, velocity components satisfy Eq. (1).

The thermo physical characteristics included in the mathematical framework are

$$W_1 = \frac{\mu_{hnf}}{\mu_f}, W_2 = \frac{\rho_{hnf}}{\rho_f}, W_3 = \frac{\sigma_{hnf}}{\sigma_f}, W_4 = \frac{(\rho c_p)_{hnf}}{(\rho c_p)_f}, W_5 = \frac{k_{hnf}}{k_f} \tag{10}$$

$$\left.\begin{aligned}
&W_1 = \frac{1}{(1-\phi_{Cu})^{2.5}(1-\phi_{Al_2O_3})^{2.5}},\\
&W_2 = \left\{(1-\phi_{Al_2O_3})\left[(1-\phi_{Cu}) + \phi_{Cu}\left(\frac{\rho_{Cu}}{\rho_f}\right)\right] + \phi_2\frac{\rho_{Al_2O_3}}{\rho_f}\right\},\\
&W_3 = (1-\phi_{Al_2O_3})\left[(1-\phi_{Cu}) + \phi_{Cu}\left(\frac{(\rho c_p)_{Cu}}{(\rho c_p)_f}\right)\right] + \phi_2\frac{(\rho c_p)_{Al_2O_3}}{(\rho c_p)_f},\\
&W_4 = \frac{k_{Cu} + 2k_{bf} - 2\phi_{Al_2O_3}(k_{bf} - k_{Al_2O_3})}{k_{Al_2O_3} + 2k_{bf} + \phi_{Al_2O_3}(k_{bf} - k_{Al_2O_3})} \times \frac{k_{Cu} + 2k_f - 2\phi_{Cu}(k_f - k_{Cu})}{k_{Cu} + 2k_f + \phi_{Cu}(k_f - k_{Cu})},\\
&W_5 = \frac{\sigma_{Al_2O_3} + 2\sigma_{nf} - 2\phi_{Al_2O_3}(\sigma_{nf} - \sigma_{Al_2O_3})}{\sigma_{Al_2O_3} + 2\sigma_{nf} + \phi_{Al_2O_3}(\sigma_{nf} - \sigma_{Al_2O_3})} \times \frac{\sigma_{Cu} + 2\sigma_f - 2\phi_{Cu}(\sigma_f - \sigma_{Cu})}{\sigma_{Cu} + 2\sigma_f + \phi_{Cu}(\sigma_f - \sigma_{Cu})}.
\end{aligned}\right\}$$

By substituting Eq. (9) into Eqs. (2) and (6), the resultant equations are:

$$\frac{\mu_{hnf}}{\mu_f}\left(1 + \frac{1}{\gamma}\right)f''' + \frac{\rho_{hnf}}{\rho_f}\left(ff'' - f'^2\right) - Mf' = 0 \tag{11}$$

$$\begin{aligned}
&\frac{k_{hnf}}{k_f}\theta'' + \left[\left(3(\theta_w - 1)\theta'^2\left(1 + \theta(\theta_w - 1)\right)^2 + 1 + \theta(\theta_w - 1)\right)^3\theta''\right]R\\
&\Pr\left(f\theta' - f'\theta\right)(\rho c_p)_{hnf}/(\rho c_p)_f + \Pr\left(Ec(f'')^2\frac{\mu_{hnf}}{\mu_f} - M\,Ec(f')^2\right) + f'A^* + \theta B^* = 0
\end{aligned} \tag{12}$$

Where $u_w = ax$, $M = \dfrac{\sigma B_0^{\,2}}{\rho_f a}$, $\Pr = \dfrac{(\mu c_p)_f}{k_f}$, $Ec = \dfrac{u_w^2}{(c_p)_{hnf}(T_w - T_\infty)}$, $R = \dfrac{16\sigma^* T_\infty^{\,3}}{3k_{hnf} k^*}$,

$Bi = \left(\dfrac{h_f}{k_f}\right)\sqrt{\upsilon_f / c}$.

$\theta_w = 1$ Represent the basic radiative case and $\theta_w > 1$ represent the radiative case which is not linear.

The related border line situations are:

$$f = s,\ f' = 1,\ \theta' = 1\ f' \to 0,\ \theta \to 0 \text{ as } n \to \infty \tag{13}$$

Where $u_w = ax$, $M = \dfrac{\sigma B_0^{\,2}}{\rho_f a}$, $\Pr = \dfrac{(\mu c_p)_f}{k_f}$, $Ec = \dfrac{u_w^2}{(c_p)_{hnf}(T_w - T_\infty)}$, $R = \dfrac{16\sigma^* T_\infty^{\,3}}{3k_{hnf} k^*}$,

$\theta_w = 1$ Represent the basic radiative case and $\theta_w > 1$ represent the radiative case which is not linear.

The dimensional form of Skin friction coefficient and Nusselt number are defined as:

Physical quantities that are worth mentioning in engineering sectors are skin friction plus Nusselt number, and are framed mathematically as:

$$C_f = \frac{2\mu_{hnf}}{\rho_{hnf} u_w^2}\left(\partial u / \partial y\right)_{y=0},\ Nu = \frac{h\left(q_r - \partial T / \partial y\, k_{hnf}\right)_{y=0}}{k_f\left(T_w - T_\infty\right)} \tag{14}$$

Employing the dimension less conversions as in Eq. (9) into Eq. (12), we obtain

$$\left.\begin{aligned} &C_f \sqrt{Re_x} = (1-\phi_1)^{-2.5}(1-\phi_2)^{-2.5}\ f''(0), \\ &\mathrm{Nu}_x = -\frac{k_{hnf}}{k_f} Re_x^{1/2}\left(R\left(1 + (\theta_w - 1)\theta(0)\right)^3 + 1\right)\theta'(0), \end{aligned}\right\} \tag{15}$$

Numerical procedure

The nature of ordinary differential equations (10 –12) along with boundary conditions (13) are highly nonlinear nature. For the determination of handling with highly nonlinear equations, we implement a numerical method identified as the bvp5c in MATLAB software.

$$f''' = -\frac{1}{W_1\left(1+\frac{1}{\gamma}\right)}\left(W_1\left(ff''-f'^2\right)-\frac{W_3}{W_2}Mf'\right) \tag{16}$$

$$\theta'' = -\frac{1}{W_5+\left[\begin{pmatrix}3(\theta_w-1)\theta'^2\left(1+\theta(\theta_w-1)\right)^2\\ +1+\theta(\theta_w-1)\end{pmatrix}^3\right]}\begin{pmatrix}Pr\left(f\theta'-f'\theta\right)W_4+Pr\begin{pmatrix}Ec\left(f''\right)^2W_1\\ -\frac{W_3}{W_4}MEc\left(f'\right)^2\end{pmatrix}\\ +f'A^*+\theta B^*\end{pmatrix} \tag{17}$$

The related border line situations are:

$$f=s,\ f'=1,\ \theta'=1\ f'\to 0,\ \theta\to 0 \text{ as } n\to\infty \tag{18}$$

The following steps are

$$f=\mathrm{H}_1,\ f'=\mathrm{H}_2,\ f''=\mathrm{H}_3,\ f'''=\mathrm{H}_3',\theta=\mathrm{H}_4,\theta'=\mathrm{H}_5,\theta''=\mathrm{H}_5' \tag{19}$$

By employing the earlier indicated substitution in a way that generates ordinary differential equations which are a preparation to first-order ordinary differential equations. The steps for solving the equations are given below.

$$\mathrm{H}_1'=\mathrm{H}_2$$

$$\mathrm{H}_2'=\mathrm{H}_3$$

$$\mathrm{H}_3'=-\frac{1}{W_1\left(1+\frac{1}{\gamma}\right)}\left[W_1\left(\mathrm{H}_1\mathrm{H}_3-\left(\mathrm{H}_2\right)^2\right)-\frac{W_3}{W_2}M\mathrm{H}_2\right]$$

$$\mathrm{H}_4'=\mathrm{H}_5$$

$$\mathrm{H}_5'=-\left[\frac{1}{W_5+\left[\begin{pmatrix}3(\theta_w-1)(\mathrm{H}_5)^2\left(1+\mathrm{H}_4(\theta_w-1)\right)^2\\ +1+\mathrm{H}_4(\theta_w-1)\end{pmatrix}^3\right](R)}\begin{pmatrix}Pr\left(\mathrm{H}_1\mathrm{H}_5-\mathrm{H}_2\mathrm{H}_4\right)W_4+Pr\begin{pmatrix}Ec\left(\mathrm{H}_3\right)^2W_1\\ -\frac{W_3}{W_4}MEc\left(\mathrm{H}_2\right)^2\end{pmatrix}\\ +\mathrm{H}_2A^*+\mathrm{H}_4B^*\end{pmatrix}\right] \tag{20}$$

along with the boundary conditions are

$$\left.\begin{aligned}&W_1(\eta)=s, W_2(\eta)=1, W_4(\eta)=1, \, at\ \eta=0\\ &W_2(\eta)\to 0, W_4(\eta)\to 0\ \ as\ \eta\to\infty.\end{aligned}\right\} \tag{21}$$

By employing the earlier indicated substitution in a way that generates ordinary differential equations which are a preparation to first-order ordinary differential equations. The steps for solving the equations are detailed in many references.

3. Plan of solution and validation

The dimensionless system of equations (10)-(12) are altered into 1st order equations via innovative dependent variables. Unknown primary conditions are carried out via shooting iteration procedure. Afterwards, calculable value for η at ∞ is honored thus and in order that all the far and away border line circumstances are pleased asymptote. Our primary simplifications are based on a range of standards at η_∞, specifically the necessity to understand the asymptotic conditions at the far field boundary for all parameter estimates. Later locale bounded value for η_∞ simplifications be carry out with $\nabla_\eta = 0.01$ and the convergences tenets 1×10^{-6} owed it 6th order abridgement fault, this scheme has gains over plentiful other numerical techniques. To test out the relevance of this current numerical procedure, we equated $-\theta'(0)$ with numerous values of *Pr* when $\gamma = 0, M = 0, \quad Ec = 0, \quad R = 0, A^* = 0, B^* = 0,$ and it processions well concurrence, is displayed in Table 2. For current investigation, the universal values for the relevant parameters are taken as $\gamma = 0.1,\ M = 0.5\ R = 1.0,\ A^* = 0.1,\ B^* = 0.1,$ Pr = 6.2, $Ec = 0.3,$

Figure 2 shows that comparison between nanofluid and hybrid nanofluid. Figure 3 impression of magnetic field parameter on the velocity profile. The velocity profile $f'(\eta)$ decreased within the boundary layer with raising magnetic parameter. As a matter of fact, the interaction of the magnetic field results in the generation of Lorentz force magnetohydrodynamic drag, which acts as a barrier to the mobility of the fluid throughout the entire domain. A Lorentz force is produced when semiconducting nanoparticles are combined with an oblique magnetic field that is put on the system. In addition, the value of the Lorentz force increases in proportion to the magnitude of the magnetic field that is being applied. This has the effect of preventing fluid mobility within the boundary layer and lowering the viscosity of the velocity boundary layer.

Figure 4 explains the effect of Casson parameter on velocity distributions of the nanoparticles. Physically, the Casson fluid parameter represents the reciprocal relationship between the yield stress and the rate of fluid viscosity. Therefore, the velocity profile decreased when higher values of the Casson fluid parameter. It is anticipated from Figure 5 that temperature profiles reveal a growing trend with increasing *M*. Physically, it appears that for growing magnetic field values the Lorentz forces rises which escalate the opposite forces to the particles of the fluid and so the temperature grows. Thermal energy is responsible for the enhancement of the Lorentz force, which is achieved by encouraging hybrid nanofluids to release sub kinetic energy. Near the boundary layer, the temperature rises as a result of the conversion of kinetic energy into heat energy, which is caused by the fluid's sluggish motion. Additionally, it may be noted that the magnetic parameter appears to function as a source of heat in the flow of the nanofluid. Figure 6 shows that the temperature profile increased when increasing values of the radiation parameter, which in turn improves the thickness of the thermal boundary layer. Radiation from heat sources raises the temperature profile because it physically alters the thermal diffusibility of the medium. Increases in thermal radiation parameters are physically associated with hotter temperatures and thicker thermal boundary layers. The temperature profile enhanced for the larger values of the *Ec* parameter, which is presented in Figure 7. Physically, enhances in the Eckert number are accompanied by an intensification of viscous dispersion, which is defined as the transformation of kinetic energy into heat energy as a result of internal frictional forces inside the medium. The fluid's temperature increases as a result of the generation of this increased thermal power. In order to investigate the characteristics of temperature for a bigger non-uniform heat source/sink parameter which are presented in Figures 8 and 9. Physically, increasing values of A* and B* represents for production of more heat. Depressive values are for heat absorbers, while affirmative values correspond to heat generators and work similarly to heat generators. In general, an increase in the heat source will result in an increase in the thermal boundary thickness, which will lead to an increased in temperature profile as seen in Figures 8 and 9. The behaviour of *Pr* on temperature is discussed in Figure 10. As *Pr* is inversely proportional to the thermal diffusivity, for greater values of *Pr*, the temperature of the fluid drops offs and is displayed in profile. Here we can analyze that for higher values of Pr, momentum diffusivity dominates over the thermal diffusivity. In short, an upturn in the Prandtl number means a deliberate amount of thermal dispersion. Figure 11. Demonstrates that the average heat transfer rate for the various values of $M\ and\ \gamma$. Increasing $M\ and\ \gamma$ decreases the average heat transmission rate. Figure 12. Display the Nusselt number rate of the nanofluids with different values of $M\ and\ R$. The Nusselt number profile increased by increasing $M\ and\ R$. Figure 13 shows that the impact of $M\ and\ A^*$ on Nusselt number profile. Higher values of A^* parameter heat transfer rate boundary layer decreased. Figure 14 represents the effect of M *and* B^* on Nusselt number profile. Increasing values of B^* parameter the Nusselt number profile declined.

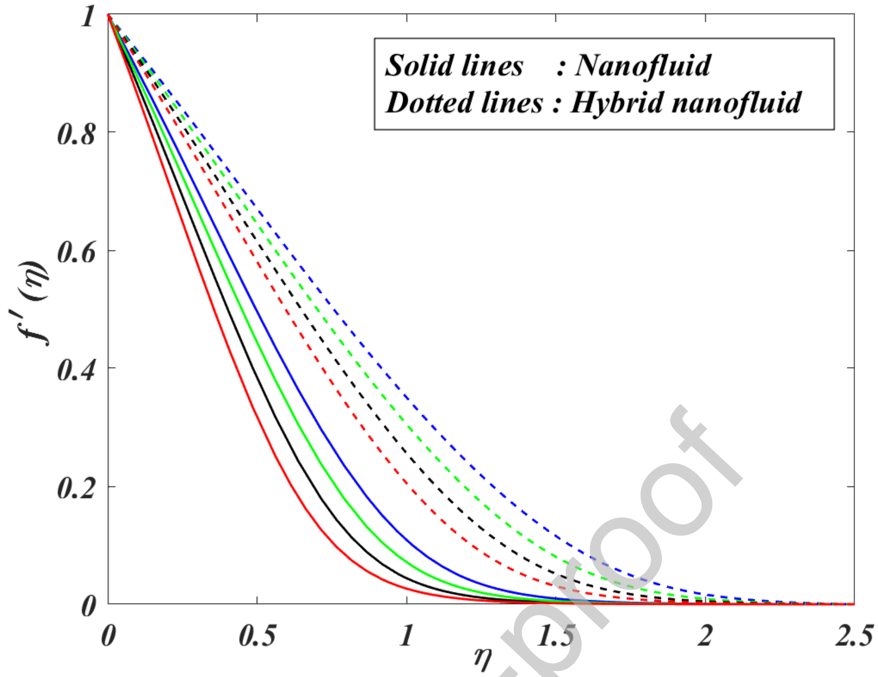

Fig 2. Comparison between nanofluid and hybrid nanofluid on $f'(\eta)$.

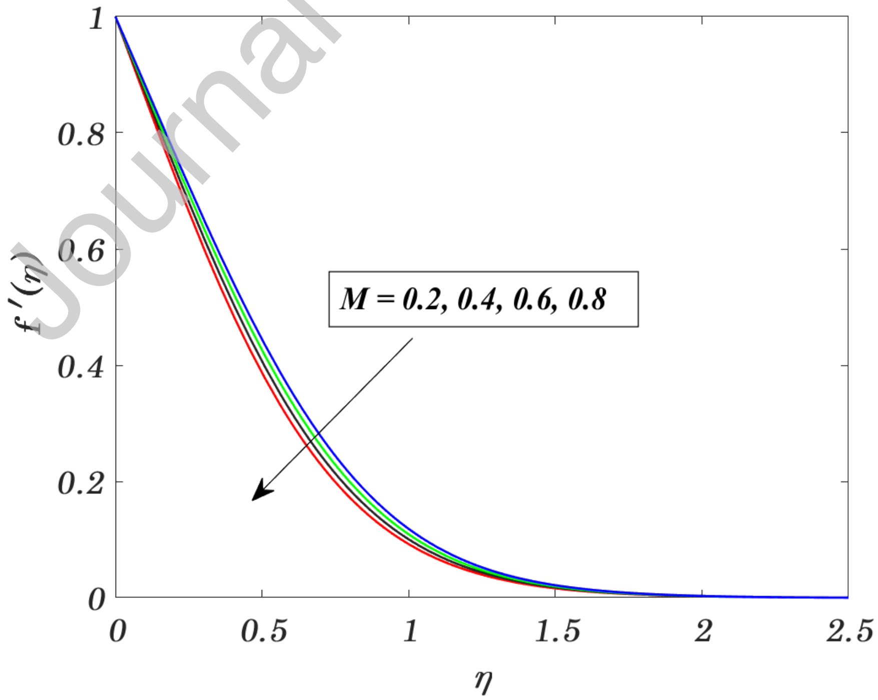

Fig 3. Impact of M on $f'(\eta)$.

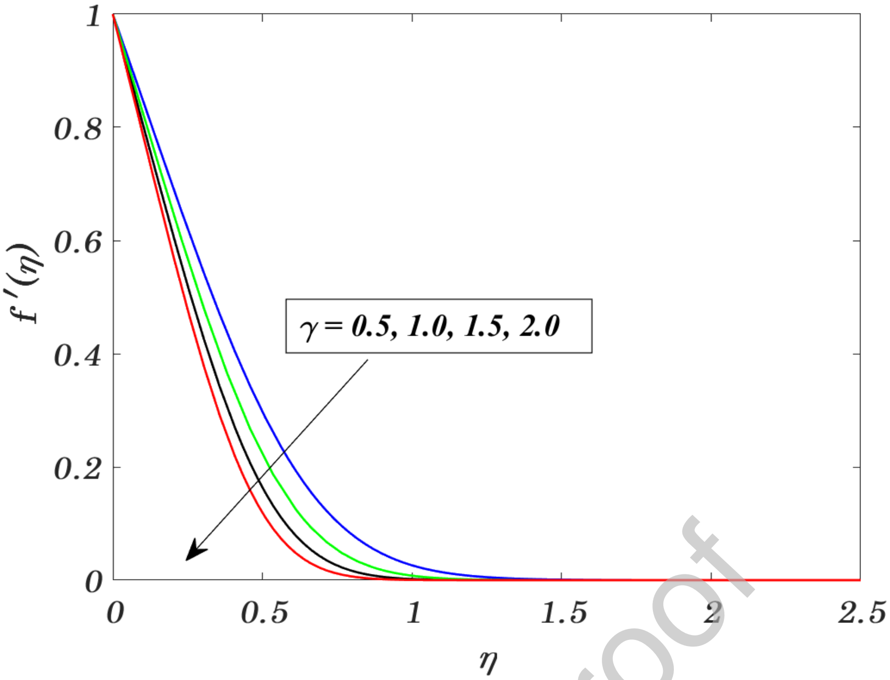

Fig 4. Impact of γ on $f'(\eta)$.

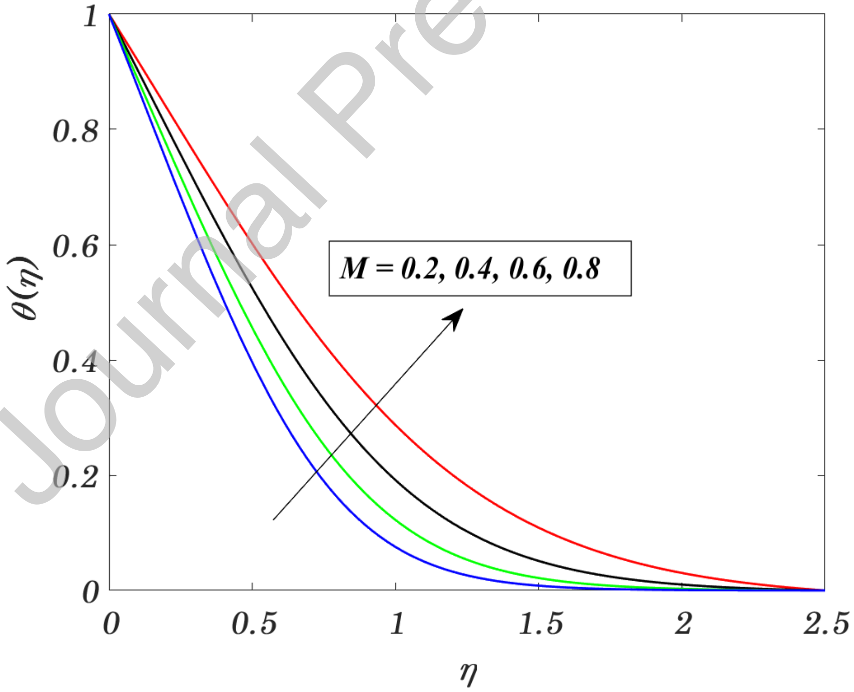

Fig 5. Impact of M on $\theta(\eta)$.

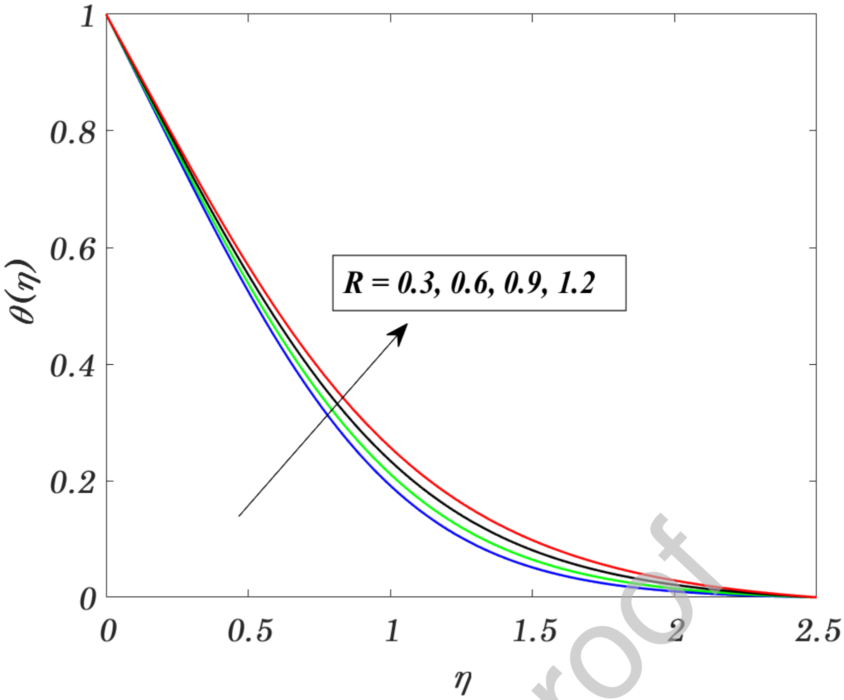

Fig 6. Impact of R on $\theta(\eta)$.

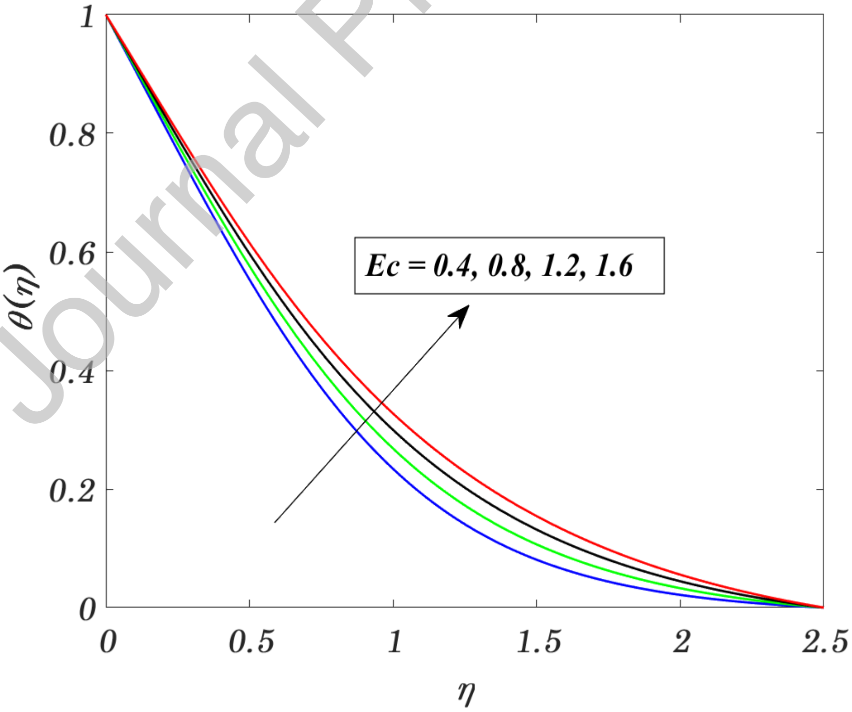

Fig 7. Impact of Ec on $\theta(\eta)$.

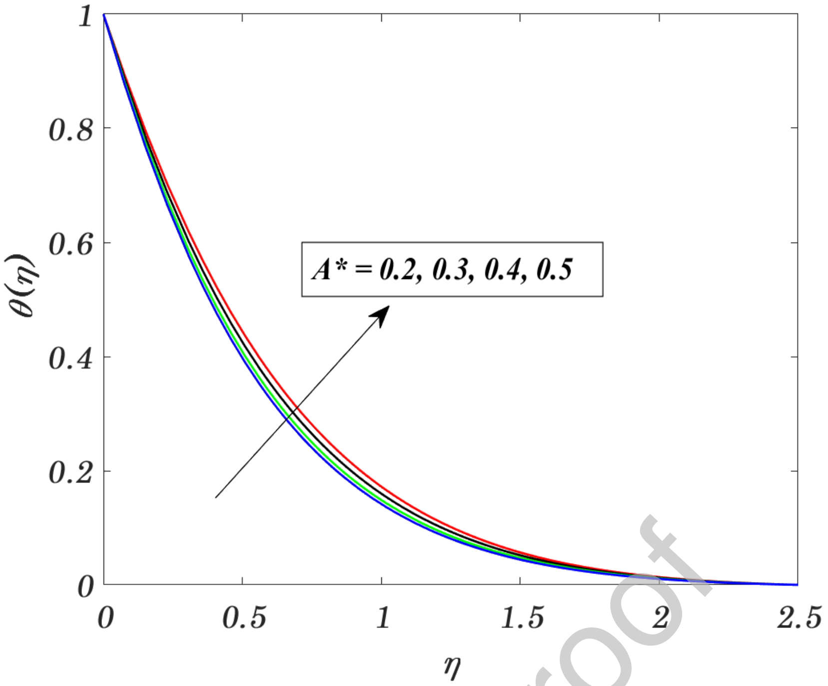

Fig 8. Impact of A* on $\theta(\eta)$.

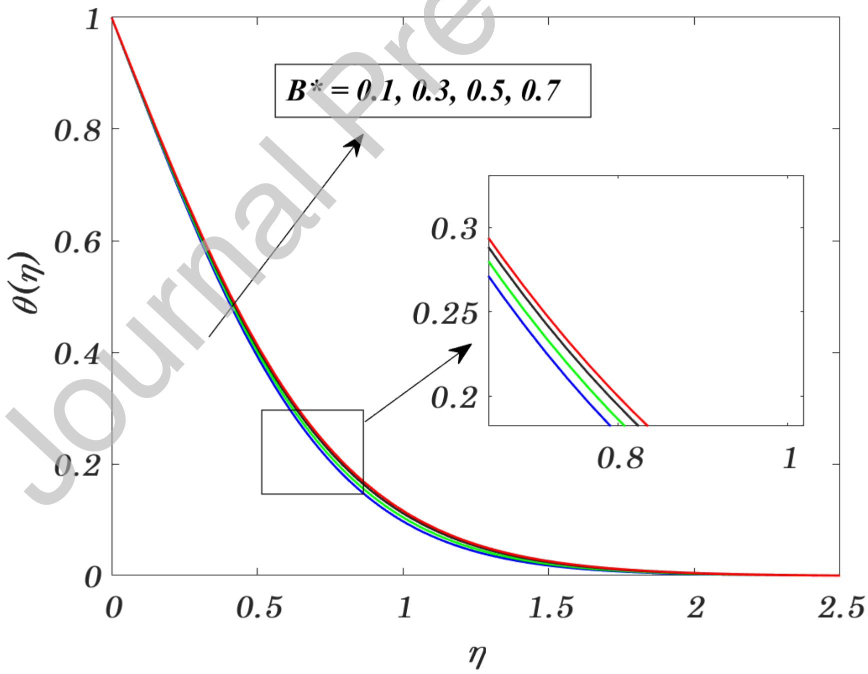

Fig 9. Impact of B* on $\theta(\eta)$.

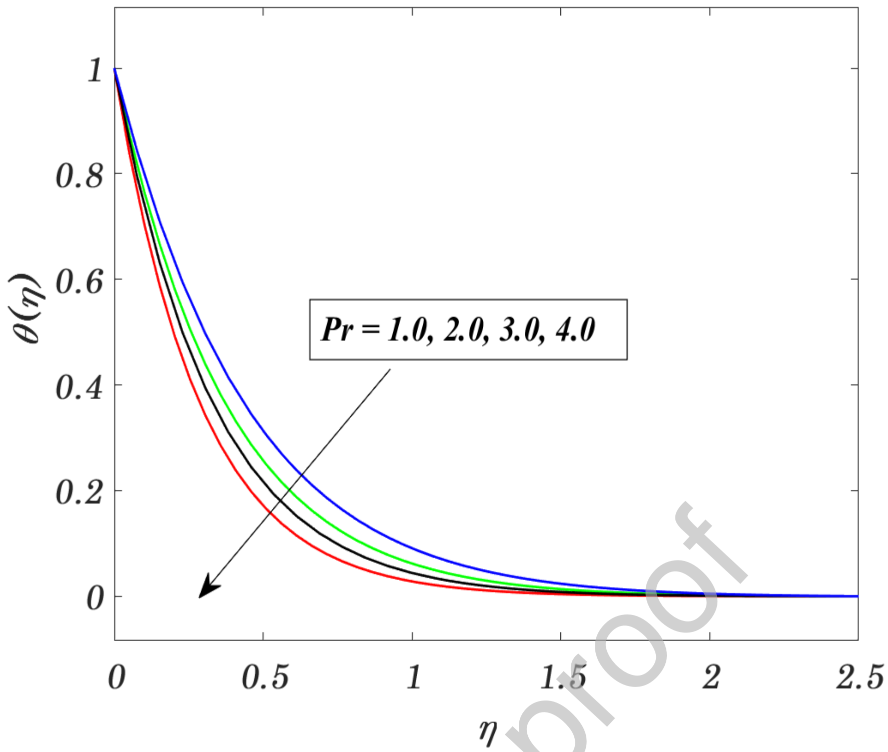

Fig 10. Impact of Pr on $\theta(\eta)$.

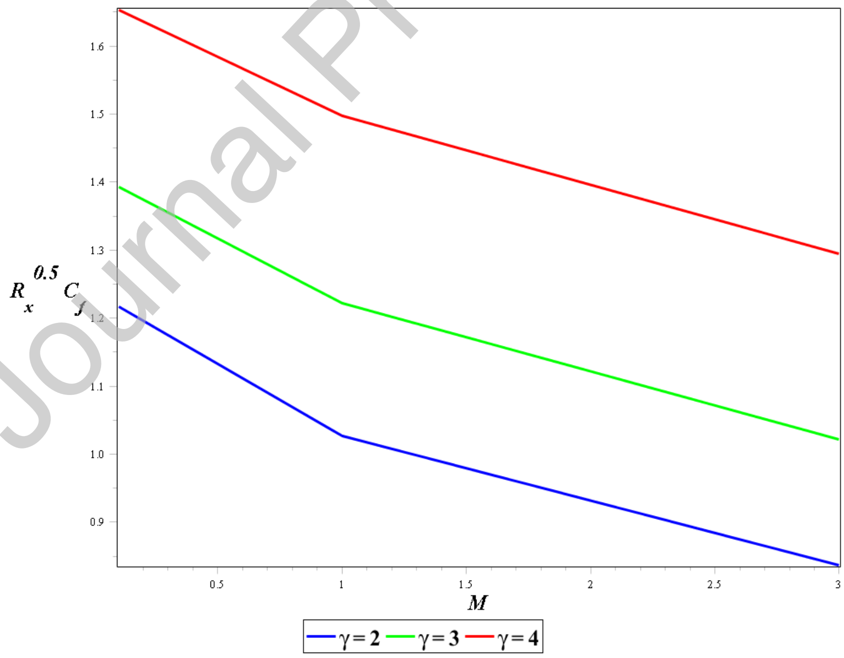

Fig 11. Impact of M and γ on $C_f R_x^{0.5}$.

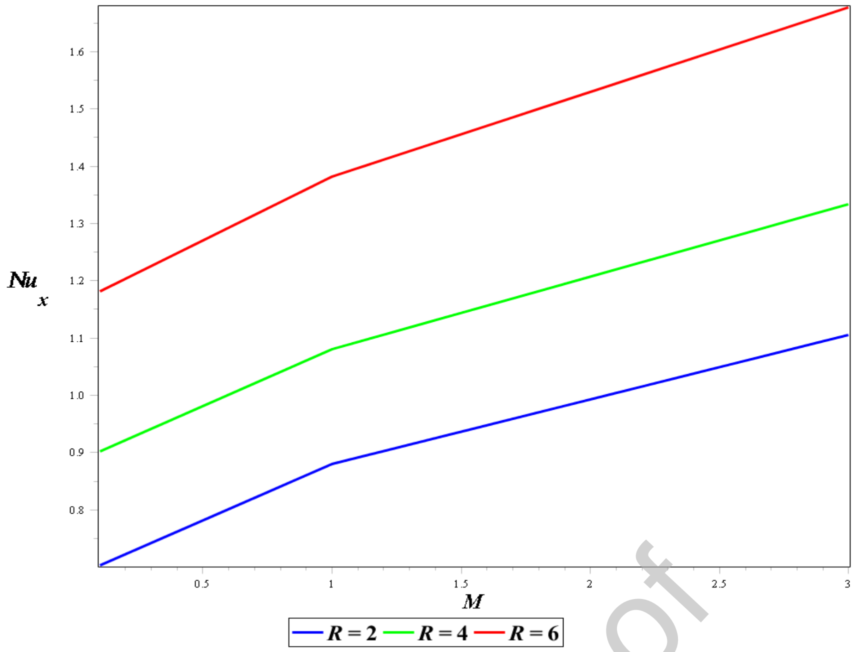

Fig 12. Impact of M *and* R on Nu_x.

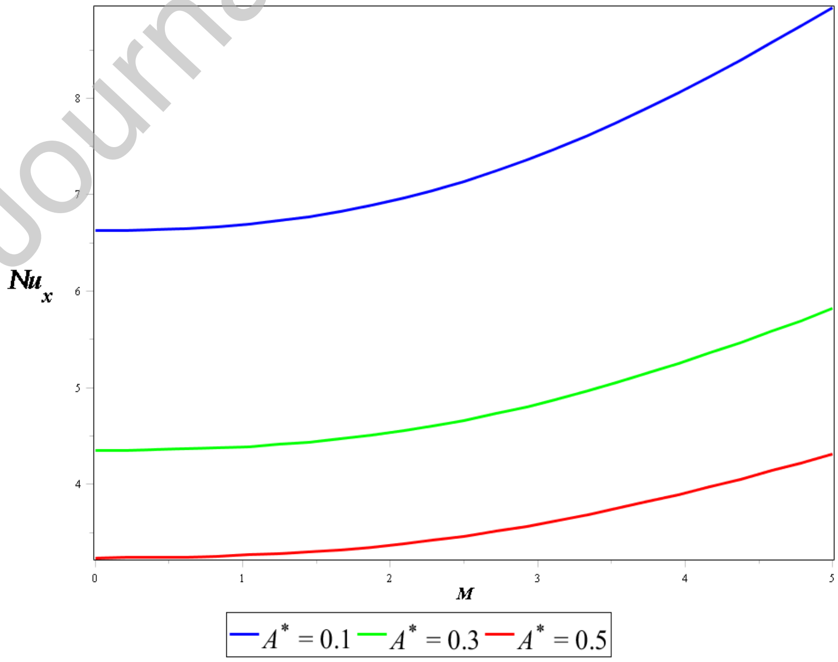

Fig 13. Impact of M *and* A^* on Nu_x.

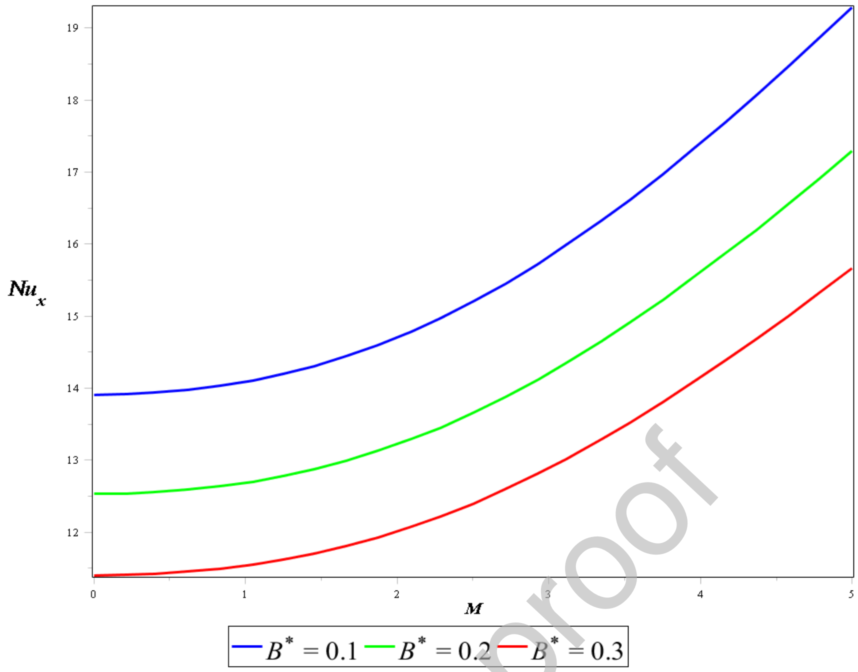

Fig 14. Impact of M *and* B^* on Nu_x.

Table 2: Comparison $-\theta'(0)$ for quite a few values of Prandtl number when $M = \ R = Ec =$ $A^* = B^* = s_f = 0.$

Pr	Mukhopadhyay and Gorla [46]	Present results
1	0.9547	0.95479
2	1.4714	1.47142
3	1.8691	1.86918
5	2.5001	2.50015
10	3.6603	3.66030

5. Concluding Remarks

The MHD Casson hybrid nanofluid with nonuniform heat source or sink has been discussed. The mathematical flow equations are established with the help of continuity, momentum and energy equations. The numerical method was applied to get exact numerical solutions to the ensuing ODEs. The primary purpose of the research is to perform an investigation into a broad range of potential uses in the fields of water purification industry, biological applications and cooling systems.

Some significant finishing observations are gathered below:

- The velocity of the hybrid nanofluid declined due to the upsurge of magnetic field, whereas a reverse behavior is observed for temperature.
- Higher values of the Casson fluid parameter the velocity profile declined.
- Increasing values of the Eckert number leads to higher temperatures.
- The energy profile increased when increasing the radiation parameter values.
- Skin friction profile enhanced with increasing the Casson fluid parameter.
- Nusselt number boundary layer increased when increasing the radiation parameter values while the opposite tendency we have noticed on A* and B* parameters cases.
- Conducting flows cooling rate can be accelerated by utilizing the Prandtl number.
- As the radiation parameter values rise, the temperature rises as well. One possible explanation for this is an increase in the effective thermal diffusivity.

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Declaration of interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.